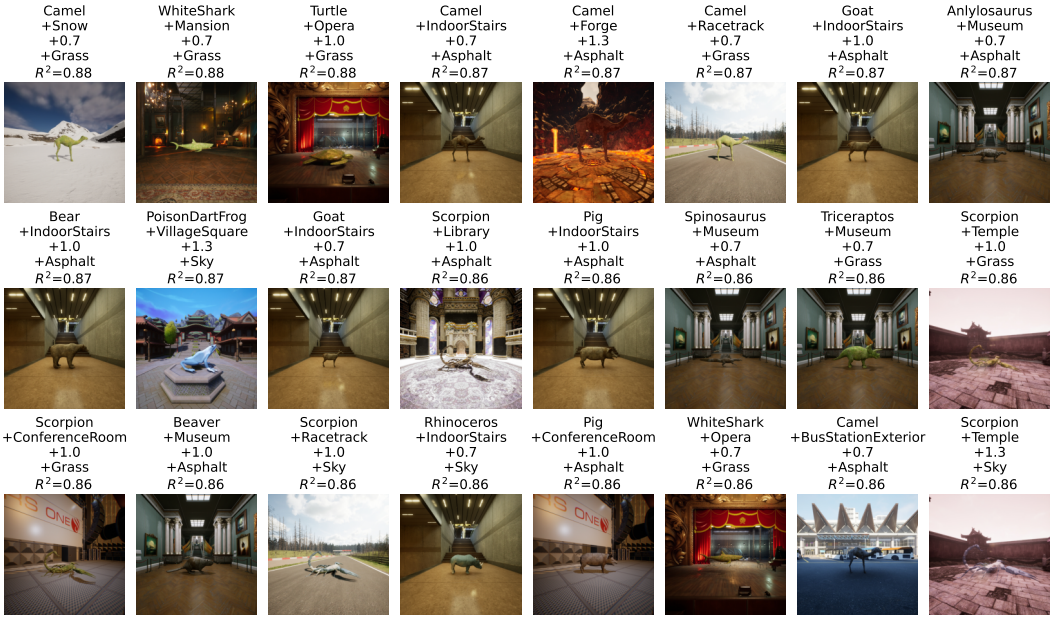
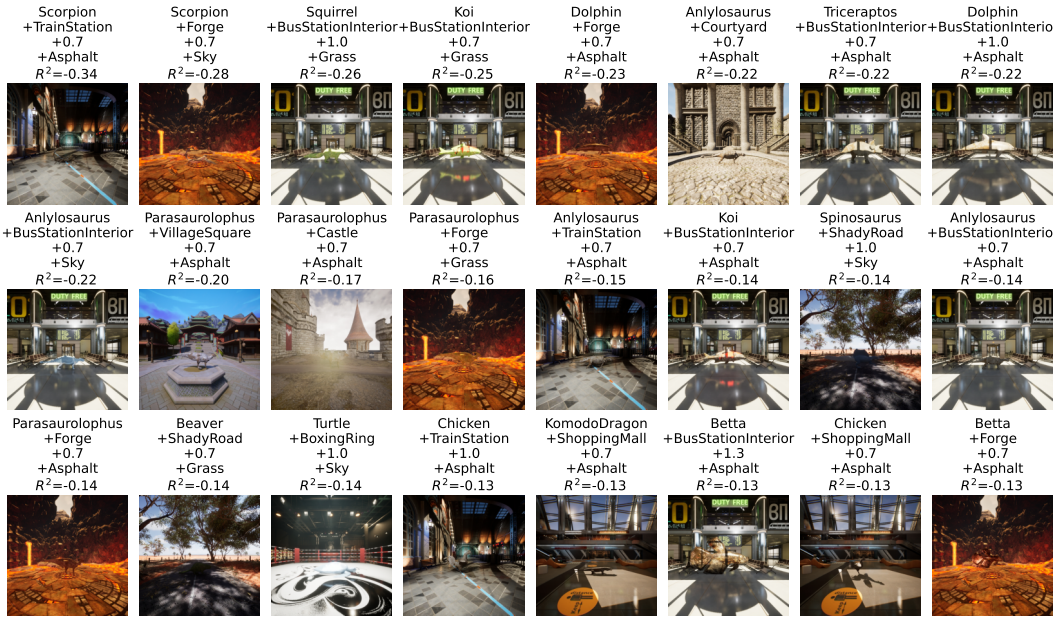


(a) Top-scoring samples in terms of R^2.



(b) Bottom-scoring samples in terms of R^2.

Figure 30. **Qualitative examples of the top- and lowest-scoring samples in PUG-Animal for the SigLIP2 model.** Each sample shows its character name, world name, size value (0.7 corresponds to "small", 1.0 corresponds to "medium", 1.3 corresponds to "large"), texture name, and its R^2 score.

I.4.2. Experiments on ImageNet-AO

We additionally perform experiments on a coarse-captioned dataset ImageNet-AO Abbasi et al. (2024), where each image sample has an associated caption composed of an adjective and a noun.

The experiments here are slightly dissimilar from the main experiments in Section 5.1, for a few reasons: (1) scarcity of per-combination data, (2) inability to train linear probes, (3) noisy/ambiguous data, and (4) coarse categories. Regardless, our framework still applies.

Dataset description. The dataset contains images described by an adjective and a noun. There are around 80 unique adjectives and over 600 unique nouns. To make the analysis balanced, we work with the dataset restricted to the most common 80 nouns and adjectives. Each potential combination of adjective and noun may have between 0 and 6 images. The dataset is thus sparse, and many of the potential combinations are not observed in the dataset. This results in a total of 3243 datapoints. We illustrate the sparsity and the pairs we work with in Fig. 31.

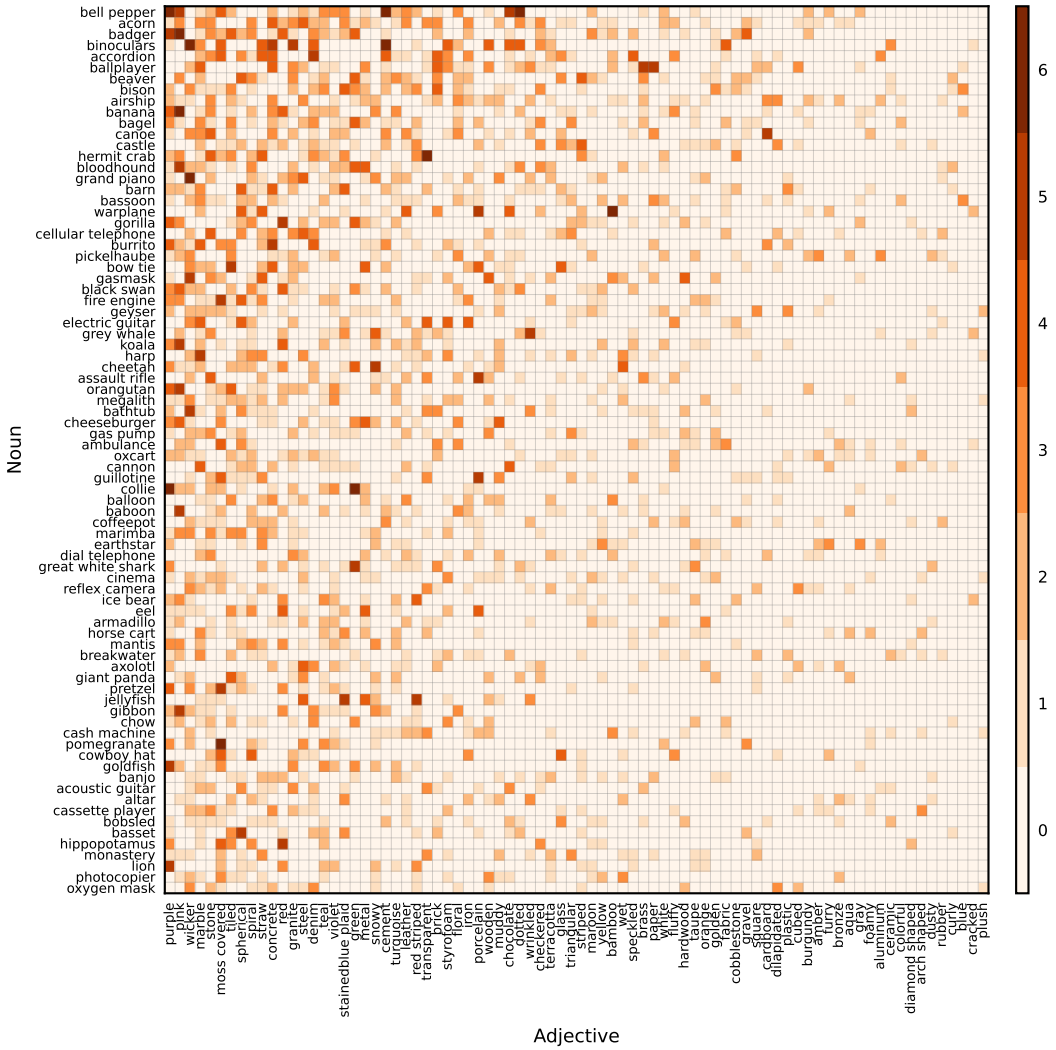


Figure 31. **Adjective-noun count matrix for ImageNet-AO (Abbasi et al., 2024) of the top 80 adjectives and nouns.** The adjective-noun pairs are sparse, and many of them are not observed in the dataset.

General setup. Due to limited availability of the data samples, we *do not* train linear probes. Because each sample is associated with a (noun, adjective) combination, we instead use the probes from the text encoder to assess the performance of the models (as detailed in the main text in Section 3.4). Concretely, we pass captions in the style of "A picture of <noun>" in the case of noun, and "A picture showing <adjective>" in the case of adjective, through the text encoder.

Because of imbalance and sparsity, we cannot rely on averaging to extract the factors as done in Section 5.1. Instead, we follow (Uselis et al., 2025) and solve a linear system of equations to recover the factors. Concretely, we construct a design matrix $\boldsymbol{A} \in \{0,1\}^{3243\times 80\cdot 2}$ where each row corresponds to a sample, and each column corresponds to either the presence of a noun (if the column index < 80) or the presence of an adjective (if the column index ≥ 80). The matrix was of full rank $2\cdot 80 - 1$. Then, we solve the linear system $\boldsymbol{A}\begin{bmatrix}\boldsymbol{u}_{\text{noun}}\\ \boldsymbol{u}_{\text{adj}}\end{bmatrix} = \boldsymbol{X}$ to recover the factors $\boldsymbol{u}_{\text{noun}} \in \mathbb{R}^{80\times d}$ and $\boldsymbol{u}_{\text{adj}} \in \mathbb{R}^{80\times d}$, where d is the dimension of the representation space, and $\boldsymbol{X} \in \mathbb{R}^{3243\times d}$ is the centered image embeddings. We show the whitened R^2 scores. The remaining procedure in the analysis follows Section 5.1.

Linearity of factors and generalization. We show the projected R^2 vs accuracy on ImageNet-AO across models in Fig. 32. As seen in

the main text (Section 5.1), higher projected R^2 coincides with higher accuracy on the full dataset. Importantly, the random baseline achieves substantially lower projected R^2 (less than 0.1) compared to the other models.

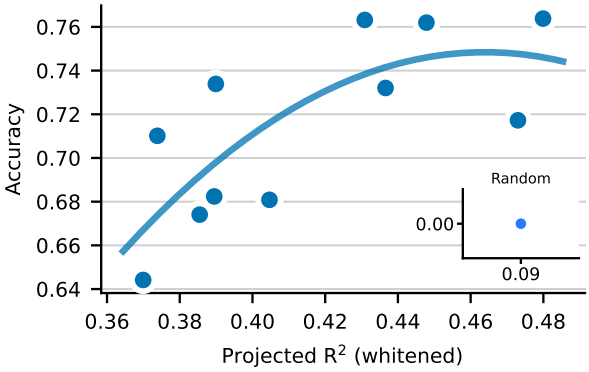


Figure 32. **Projected R^2 vs accuracy on ImageNet-AO across models.** Higher projected R^2 coincides with higher accuracy on the full dataset. Linear probes were not trained here, and the results are computed using the text encoder.

Orthogonality of the factors. To substantiate the claims of orthogonality of factors across concepts, we extract the factors for all the models as detailed in the setup above. Concretely, for each of the attribute factor $\boldsymbol{u}_i, i \in [80]$ and noun factor $\boldsymbol{u}_j, j \in [80]$, within- and across-concept orthogonality as detailed in Section 5.1.

We illustrate the results in Fig. 33. For all of the evaluated models the same pattern of orthogonality is observed: the factors are more orthogonal across concepts than they are within concepts. For example, for the CLIP ViT-L/14 model, the within-concept similarity on average is 0.10 between nouns, and 0.14 between adjectives, while the average cosine similarity across concepts is 0.07. The random baseline on average yields 0.49 cosine similarity both across and within concepts.

Interestingly, all of the non-random models exhibit surprising degree of similarity in terms of the cosine similarities. For example, CLIP ViT-L/14 and OpenCLIP ViT-L/14 on average exhibit almost the same cosine similarity within and across concepts, differing only in the noun-noun cosine similarity (0.10 vs 0.11, respectively). These results support the notions of universality between models as argued by the Platonic Representation Hypothesis (Huh et al., 2024), and empirically observed in Universal Sparse Autoencoders (Thasarathan et al., 2025).

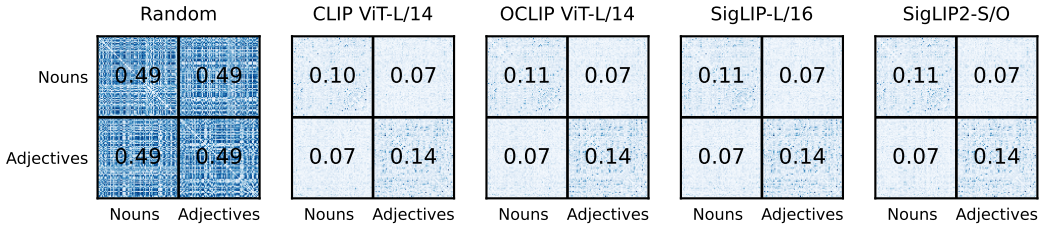


Figure 33. **Orthogonality of the factors on ImageNet-AO.** We show the cosine similarity of the factors for the SigLIP2 model on ImageNet-AO; we separate the first concept (nouns) from the second concept (adjectives) and show average similarity across each 2×2 block. The factors are more orthogonal across concepts than they are within concepts. The random baseline does not show this pattern.

Qualitative examples. To understand the results deeper, we show the qualitative examples of the top- and lowest-scoring samples in ImageNet-AO for the SigLIP2 model in Fig. 34. The top-scoring samples show high degree of projected R^2 scores (generally > 0.75), and correctly depict the adjective and noun of the sample. Even there, however, some samples are incorrectly predicted by the model, suggesting a potential lack of alignment between the image and text encoders[5].

The lowest-scoring samples show low degree of projected R^2 scores (generally < 0.10), and are often incorrectly predicted by the model. Few of the samples appear to be incorrectly labeled (e.g. first image depicting a orangutan as a gorilla), while some are correctly classified by the model but show a lack of factorization.

[5]This was less of an issue in the main experiments because the image embeddings were analyzed using linear probes.